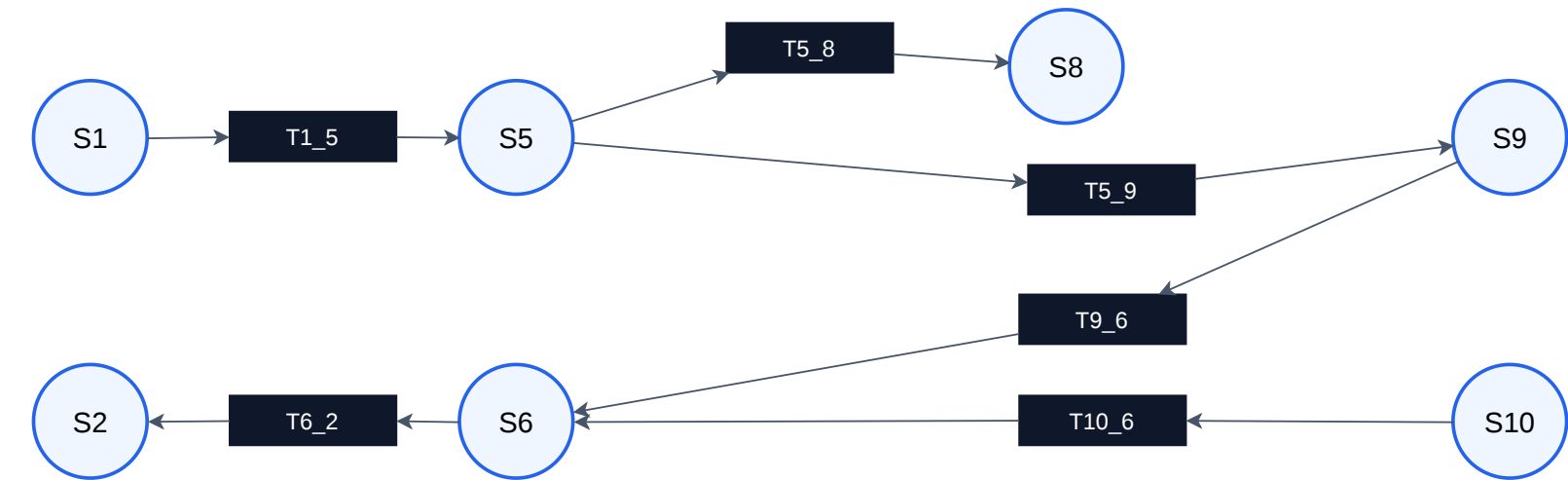


RAILWAY INTERLOCKING | PETRI NET v1

Initial design - route skeleton and reusable safety subnets | 5 October 2026

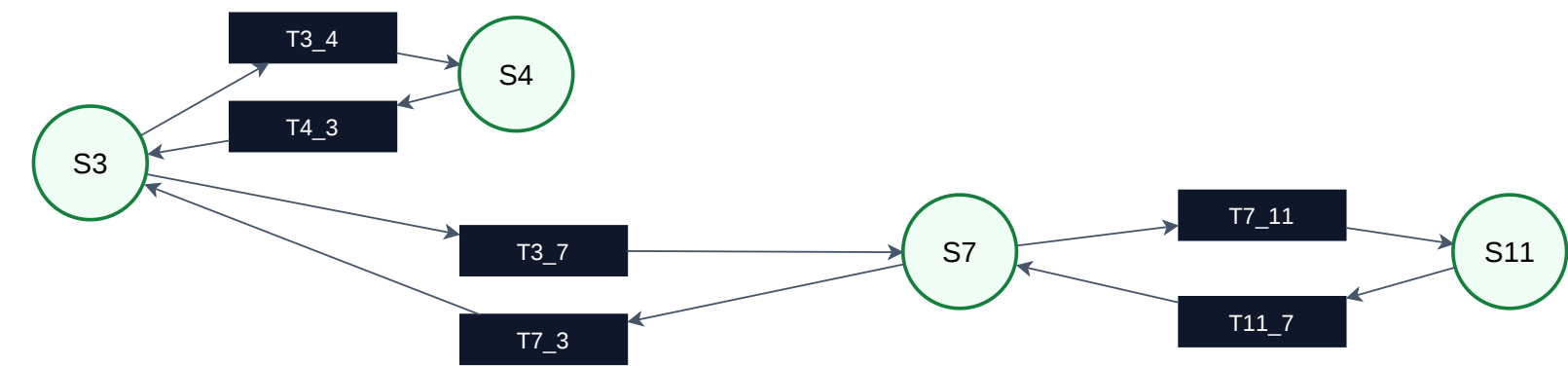
1. Passenger movement subnet



Southbound: 1 -> 5 -> 8 or 9. Northbound: 9 or 10 -> 6 -> 2.

S1...S11 hold train tokens (name, direction, destination). Every T is expanded using subnet 3.

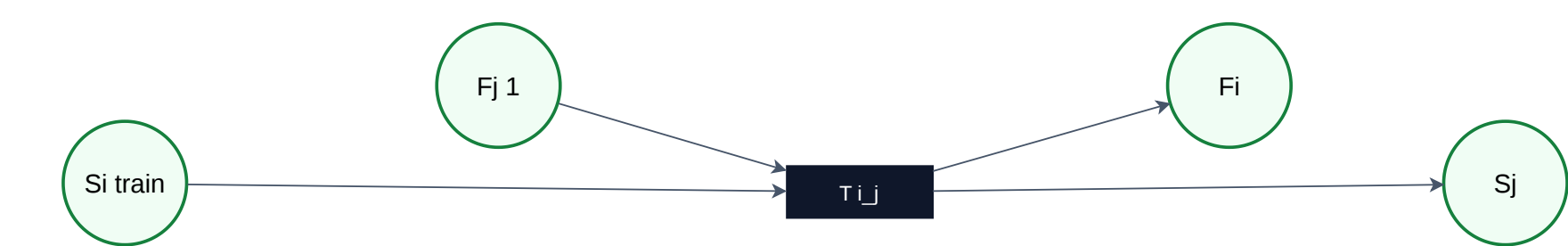
2. Freight movement subnet



Entry/destination routes: 3 -> 4; 3 -> 7 -> 11; 4 -> 3; 11 -> 7 -> 3.

No arcs connect passenger places to freight places. Initial marking: all S places are empty.

3. Expand EVERY movement with destination capacity



For each section k: Sk = occupied, Fk = free. Initial marking F1...F11 = 1; S1...S11 = 0.

Invariant: tokens(Sk) + tokens(Fk) = 1. T i_j fires only for the correct route/direction.

Entry: consume Fentry, produce Sentry. At destination: NEXT move consumes Sdest and returns Fdest.

4. Junction resources and passenger priority



J1 shared by T1_5, T6_2, T3_4 and T4_3. J2 shared by T5_8, T5_9, T9_6 and T10_6. Each resource starts with 1 token; subnet applies to each listed T.

Batch rule: choose a conflict-free set using the same pre-move marking. Each junction may be used once per moveTrains call. Hold its token until the chosen batch finishes.

Priority guard on T3_4 / T4_3: no enabled passenger crossing at J1 in the requested batch. An enabled passenger requires a free next section.

v1 scope: occupancy, route separation and conservative junction exclusion. Next review: entry/exit guards, route validation, batch scheduling and deadlock cases. This is a design draft, not a deadlock-freedom proof.